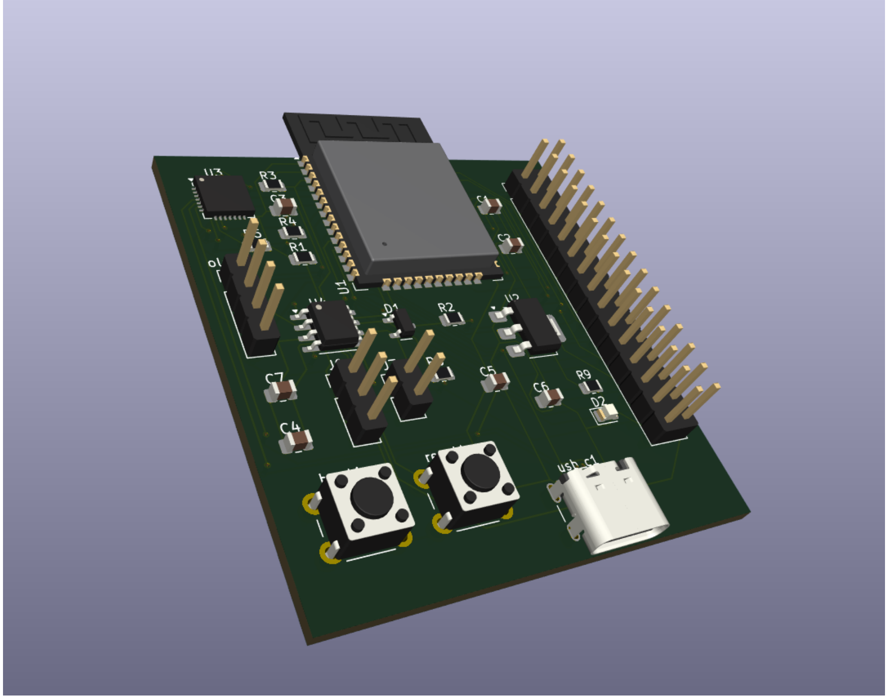
ESP32 DEVELOPMENT BOARD

Technical Datasheet

Custom 2-Layer Embedded Development Platform



Core module: ESP32-WROOM-32E | **EDA:** KiCad 9 | **PCB:** 2-layer

Portfolio / learning design

1. Product Overview

The ESP32 Development Board is a custom 2-layer embedded-development PCB built around the **ESP32-WROOM-32E**. It combines USB connectivity, regulated 3.3 V power, GPIO expansion, OLED support, UART, SPI, I²C and CAN communication in one development platform.

The project follows a complete PCB design workflow from schematic capture and component selection through footprint assignment, PCB placement, routing, design-rule verification and 3D visualization.

Key Features

Area	Implementation
MCU / wireless	ESP32-WROOM-32E module with integrated Wi-Fi and Bluetooth capability
USB	USB-C connector with CP2102N USB-to-UART bridge
Power	USB-C input with AMS1117-3.3 regulation
Display	Dedicated OLED connector
CAN	SN65HVD230 transceiver, NUP2105L protection and selectable 120 Ω termination
Expansion	30-pin GPIO header with GPIO, 3V3, 5V and GND
Controls	Boot and Reset push buttons
Indicator	User LED with 330 Ω series resistor
PCB	2-layer routed PCB design

2. System Architecture

The design is divided into functional blocks that keep the microcontroller, power, USB, display, CAN and expansion functions clearly organized.

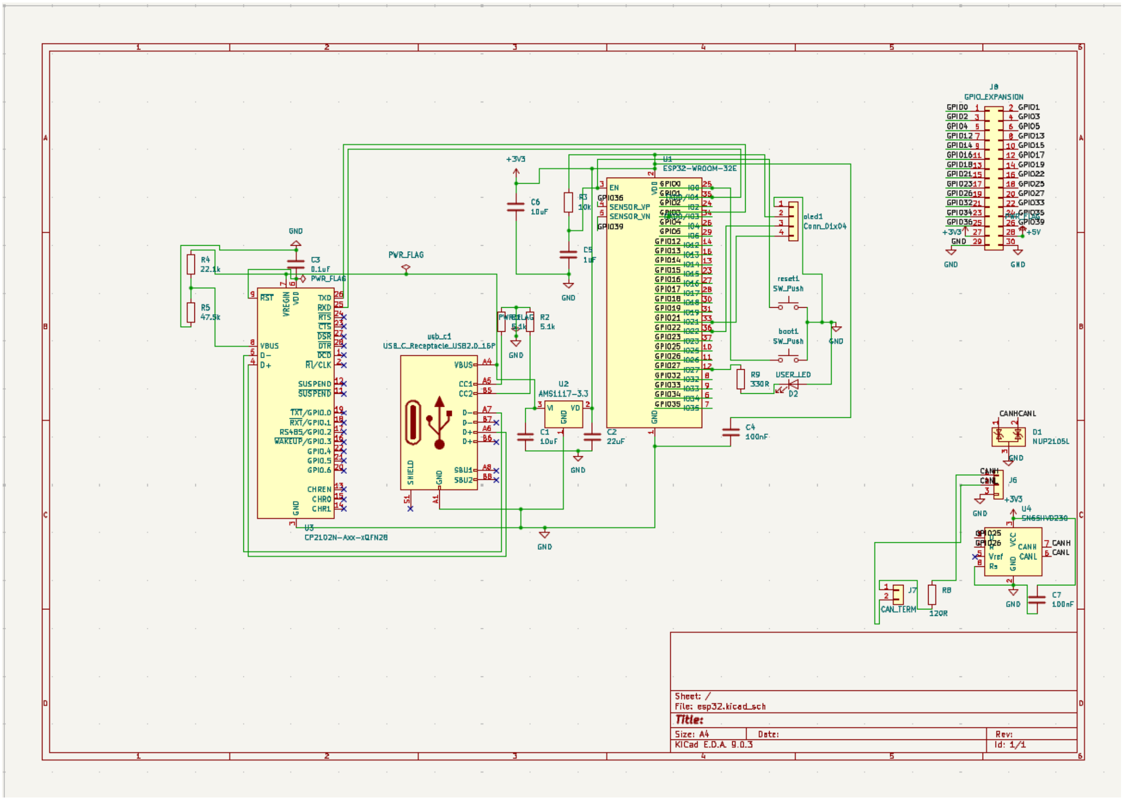
Block	Primary devices / connections	Purpose
USB / programming	USB-C → CP2102N → ESP32 UART0	Host connection, serial communication and programming path
Power	USB-C VBUS → AMS1117-3.3	Generates the board 3.3 V rail
Microcontroller	ESP32-WROOM-32E	Main processing and wireless subsystem
OLED	ESP32 I²C → OLED connector	Display/peripheral expansion
CAN	GPIO25/26 → SN65HVD230 → CAN connector	CAN bus physical-layer interface
CAN protection	NUP2105L	Protection for CANH / CANL
Termination	J7 + R8 = 120 Ω	Selectable CAN termination
Expansion	J8, 30-pin header	General-purpose GPIO and power access

Design intent

• Provide one reusable ESP32 platform for common embedded interfaces.

• Expose a broad GPIO set while retaining dedicated peripheral connectors.

• Maintain a compact 2-layer implementation for a portfolio-level PCB design exercise.



Completed schematic overview.

3. Interfaces and Connector Functions

Interface	Implementation	Notes
USB-C	USB-C receptacle	Power input and USB data connection to CP2102N
UART	CP2102N ↔ ESP32 UART0; UART2 GPIO access	UART0: GPIO1/GPIO3; UART2: GPIO16/GPIO17
I^2C	GPIO21 / GPIO22	Used for OLED and available for external I^2C peripherals
SPI	GPIO18 / GPIO19 / GPIO23	Common SPI signal set exposed through GPIO expansion
CAN	SN65HVD230 + J6	CANH, CANL and GND external bus connection
GPIO	J8, 2×15 header	26 GPIOs plus 3V3, 5V and two GND positions
OLED	Dedicated 4-position connector	Dedicated display/peripheral connection

On-board controls and indicator

• **BOOT:** push-button connected to ESP32 GPIO0 for boot-mode control.

• **RESET:** push-button connected to the ESP32 EN/reset circuit.

• **USER LED:** GPIO2 drives the user LED through R9 = 330 Ω.

EN / Reset circuit

• R3 = 10 kΩ pull-up from 3V3 to EN.

• C5 = 1 µF from EN to GND.

• Reset push-button provides an EN-to-GND reset action.

4. GPIO Expansion Header — J8

J8 is a 30-pin, 2.54 mm-pitch vertical header providing a universal expansion point for the available ESP32 GPIOs plus power and ground. GPIO6–GPIO11 are reserved for the module's internal flash interface.

Pin	Signal	Primary use / note
1	GPIO0	BOOT / general GPIO
2	GPIO1	UART0 TX / general GPIO
3	GPIO2	User LED / general GPIO
4	GPIO3	UART0 RX / general GPIO
5	GPIO4	General GPIO
6	GPIO5	SPI chip-select / general GPIO
7	GPIO12	General GPIO
8	GPIO13	General GPIO
9	GPIO14	General GPIO
10	GPIO15	General GPIO
11	GPIO16	UART2 RX / general GPIO
12	GPIO17	UART2 TX / general GPIO
13	GPIO18	SPI SCLK / general GPIO
14	GPIO19	SPI MISO / general GPIO
15	GPIO21	I²C SDA / general GPIO
16	GPIO22	I²C SCL / general GPIO
17	GPIO23	SPI MOSI / general GPIO
18	GPIO25	CAN TX / general GPIO
19	GPIO26	CAN RX / general GPIO
20	GPIO27	General GPIO
21	GPIO32	General GPIO
22	GPIO33	General GPIO
23	GPIO34	Input-only GPIO
24	GPIO35	Input-only GPIO
25	GPIO36	Input-only GPIO
26	GPIO39	Input-only GPIO
27	+3V3	3.3 V rail
28	+5V	5 V rail
29	GND	Ground
30	GND	Ground

GPIO note: GPIO34, GPIO35, GPIO36 and GPIO39 are input-only. GPIO1 and GPIO3 are shared with the USB-to-UART/programming path, while GPIO0 is used by the boot control circuit.

5. Power and USB Subsystems

The board is powered through the USB-C connector. USB VBUS feeds the power section, where the **AMS1117-3.3** regulator generates the 3.3 V rail used by the ESP32 and supporting circuitry.

Documented power and decoupling components

Reference	Value	Function
U2	AMS1117-3.3	3.3 V linear regulator
C1	1 µF	Regulator input-side capacitance
C2	22 µF	Regulator output-side bulk capacitance
C4	100 nF	ESP32/local 3.3 V decoupling
C6	10 µF	Local bulk decoupling
C7	100 nF	CAN transceiver local decoupling

USB-to-UART

The CP2102N provides the USB-to-UART bridge between the host computer and the ESP32. ESP32 UART0 uses GPIO1 (TX) and GPIO3 (RX). UART2 is available through GPIO16 and GPIO17 on the expansion header.

6. CAN Interface

The CAN interface uses an **SN65HVD230** transceiver connected to the ESP32 through GPIO25 and GPIO26. The external bus is presented at J6 as CANH, CANL and GND.

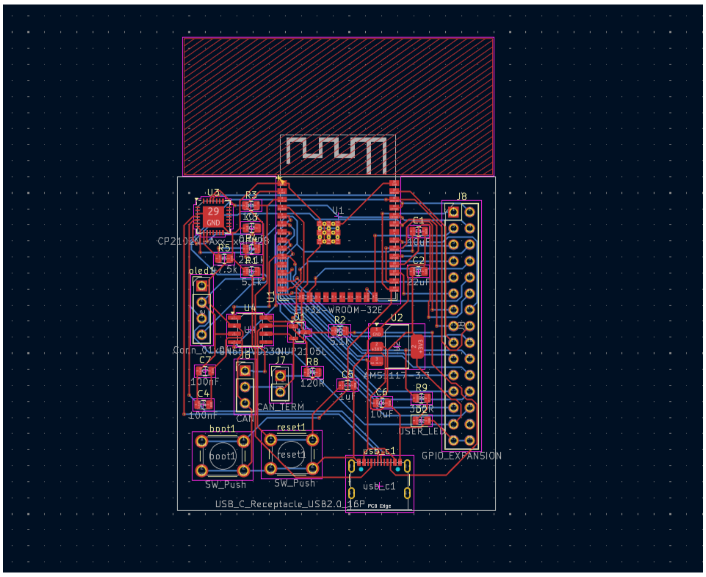
Reference	Device / value	Function
U4	SN65HVD230	CAN physical-layer transceiver
U5	NUP2105L	CANH/CANL protection device
R8	120 Ω	CAN termination resistor
J7	CAN_TERM	Selectable termination jumper
J6	CAN connector	CANH / CANL / GND external bus connection
C7	100 nF	Local transceiver decoupling

Termination behavior

When the termination jumper is installed, the 120 Ω resistor is connected between CANH and CANL, providing selectable bus termination.

Transceiver configuration

The SN65HVD230 RS pin is tied to GND for high-speed operation. The Vref pin is left floating in the design.



Completed PCB routing view.

7. PCB Implementation and Design Verification

The final PCB is a **2-layer** design. The ESP32 module is placed with its antenna extending toward the board edge and the antenna region kept clear of the main component and routing area. USB-C is positioned at the board edge, while the GPIO expansion header is placed along the right edge for convenient access.

Routing and verification

• The PCB connections are fully routed.

• The routed design was imported into KiCad through a Specctra session.

• KiCad design-rule verification was completed successfully.

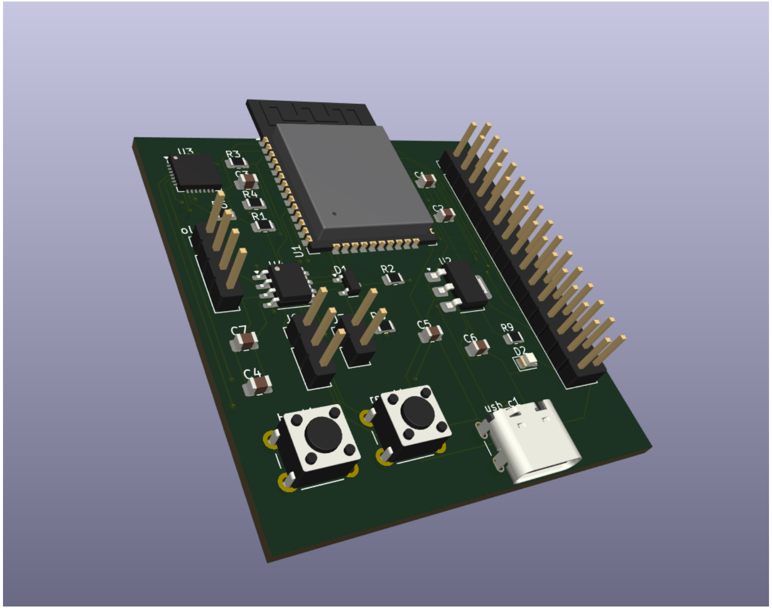
• All required connections are present.

Verification Summary

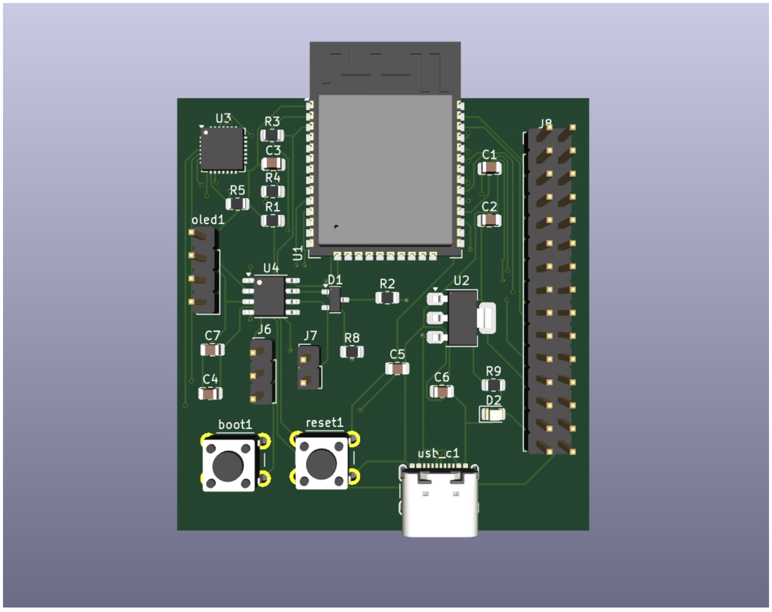
Verification item	Result
Electrical Rules Check (ERC)	PASSED
PCB Design Rules Check (DRC)	PASSED
Unconnected items	0
Unrouted connections	0

8. Visual Documentation

3D Perspective View



3D Top View



9. Project Files and Documentation

Recommended repository structure

```
ESP32-Development-Board-PCB/
■■■ README.md
■■■ esp32.kicad_pro
■■■ esp32.kicad_sch
■■■ esp32.kicad_pcb
■■■ datasheet/
■   ■■■ ESP32_Development_Board_Technical_Datasheet.pdf
■■■ images/
    ■■■ 3d-angle.png
    ■■■ 3d-top.png
    ■■■ pcb-layout.png
    ■■■ schematic.png
```

Project status

PCB Design Completed

The schematic, PCB layout, routing and design verification have been completed. The repository contains the KiCad design files, visual documentation and this technical datasheet.

10. Design Information

This datasheet documents the architecture, interfaces, pin assignments, PCB implementation and design verification of the ESP32 Development Board.

Design scope

- ESP32-WROOM-32E based embedded development platform
- USB-C programming and power interface
- USB-to-UART bridge using CP2102N
- 3.3 V regulation using AMS1117-3.3
- CAN interface with SN65HVD230
- CAN protection and selectable 120 Ω termination
- OLED interface
- UART, SPI and I²C peripheral access
- 30-pin GPIO expansion header
- Boot, Reset and User LED controls

Document information

Item	Value
Document	ESP32 Development Board — Technical Datasheet
EDA	KiCad 9
Routing assistance	FreeRouting v2.3.0
PCB stack	2-layer
Core module	ESP32-WROOM-32E
Project type	Portfolio / learning PCB design
Design status	Completed
Verification	ERC Passed • PCB DRC Passed • 0 unconnected • 0 unrouted